

Product Specification Document

SmartAir – Connected Air Quality Monitor

1. Product Overview

SmartAir is an IoT-based air quality monitoring device designed to measure indoor environmental conditions and provide real-time insights to users. It helps improve health and comfort by tracking pollutants and suggesting actions.

2. Objectives

- 1 Monitor indoor air quality in real time
- 2 Detect harmful pollutants and alert users
- 3 Provide actionable recommendations
- 4 Enable remote monitoring via mobile app

3. System Components

- 1 Air Quality Sensors (CO₂, PM_{2.5}, VOC)
- 2 Microcontroller Unit (MCU)
- 3 Wireless Connectivity Module (Wi-Fi/Bluetooth)
- 4 Mobile Application
- 5 Cloud Backend for data storage and analytics

4. Key Features

- 1 Real-time monitoring dashboard
- 2 Air quality alerts and notifications
- 3 Historical data visualization
- 4 Multi-device support

5. Technical Architecture

SmartAir uses embedded sensors connected to an MCU. Data is transmitted securely to a cloud platform via Wi-Fi. Users access insights through a mobile application communicating with REST APIs.

6. Security & Compliance

- 1 TLS encryption for all communications
- 2 Secure device authentication

- 3 GDPR-compliant data handling

7. Use Cases

- 1 Home users monitoring air quality
- 2 Offices ensuring healthy work environments
- 3 Schools tracking classroom air conditions